KANAK CHAUDHARI

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PROFESSIONAL SUMMARY

Python Backend Engineer with 4+ years of experience building and running production services in Python, alongside Go and Ruby. Builds async APIs and services with FastAPI, Starlette, Django, Flask, and asyncio, and owns them in production: process management, containers, observability, and incident response. Strongest in databases, queues, caching, and performance work. Author of LiSA, an open-source Python desktop app with nearly 200 GitHub stars, pyqtgraph contributor, and MumPy 2024 speaker.

TECHNICAL SKILLS

Python: asyncio, aiohttp, FastAPI, Starlette, Django, Django REST Framework, Flask, Sanic, Pydantic, WebSockets
Python Tooling: Gunicorn, Uvicorn, PyInstaller, PyQt5, Matplotlib, mitmproxy, Locust, boto3
Other Languages: Go, Ruby (Rails), JavaScript, C, Bash
Data Stores: PostgreSQL, MySQL, Redis, Cassandra, MongoDB, Elasticsearch, SQLite
Queues and Async: RabbitMQ (quorum queues, dead-lettering), Redis, Sidekiq
Infrastructure: Linux, Docker, Kubernetes, Jenkins, Ansible, Nginx, AWS, DigitalOcean, Vault
Observability: ELK, APM, Prometheus, Grafana, PMM, Pyroscope
Languages Spoken: English, Japanese, Hindi

PROFESSIONAL EXPERIENCE

Full Stack Software and Infrastructure Developer — PriceLabs **09/2023 - Current**

– Python Services and APIs:
- Productionized the PriceLabs MCP server (Python, MCP SDK, streamable HTTP with SSE): moved it from a single process to Gunicorn with Uvicorn workers and graceful restarts, and led its security and architecture audit (Snyk scan, external pentest, access review).
- Built an internal TLS fingerprint-spoofing proxy in Python (Flask) that handles millions of scraper requests per day, and raised its throughput 2.5x with coroutine-based concurrency.
- Built live on-demand scraping APIs for Airbnb and Vrbo metadata, length-of-stay pricing, and occupancy, used by the data science and support teams for re-scrapes and walked through with each team.
- Load-tested the new RabbitMQ HA cluster with Locust before cutover, and self-hosted an OpenStreetMap geocoder used by the scrapers as a reverse-geocoding fallback, avoiding third-party rate limits.

– Backend, Data, and Performance:
- Built Underlease, the internal platform engineers use to provision and track cloud resources (Go, HTMX, SQLite), from PRD and schema design to production, with role-based access, per-team cost quotas, and action logs; it cut about $600K in cloud costs.
- Refactored the scraping data layer behind 13 million+ listings: cut database connections from 400 to 4 per process and added prepared statements with batch inserts, making scraping 6x faster.
- Created a command-driven job system with Redis, Sidekiq, and Rails that cut system load by 5x, and sped up API responses by removing authentication database calls and moving sessions to Redis.
- Diagnosed write stalls on the pricing database: found type-mismatched query bindings forcing full table scans, added spatial indexes, and retuned redo, I/O, and isolation settings, cutting lock waits by 99%.
- Redesigned billing job execution into smaller controlled batches, reducing peak memory by 30%, and rolled out continuous profiling (Pyroscope) that traced about 78% of worker CPU to logging and JSON serialization.

– Queues, Reliability, and Team:
- Architected and ran the RabbitMQ migration from a single node to an HA quorum cluster, introduced dead-lettering across 14 queues to end poison-message loops, and moved Redis to a Sentinel setup.

- Built the database backup and restore pipeline (XtraBackup, Docker) with a restoration API and metrics exporter, and migrated about 1.5 PB of object storage to Backblaze B2.
- Created Jenkins pipelines for staggered releases and rollback and a dependency vulnerability (SCA) pipeline; lead incident response, review pull requests across codebases, interview candidates, and onboard new engineers.

IT Engineering Intern — Plus W Inc, Japan **01/2025 - 02/2025**

- Designed an event-driven architecture with Redis and Python service workers for real-time data processing.
- Developed REST APIs with Django REST Framework, with documentation and security best practices.
- Built a Chrome extension simulator for IoT devices that cut testing time by 40%.

DevOps and Security Developer — Data Cloud Gaze **10/2024 - 02/2025**

- Created a PostgreSQL containerization tool with plugin extensibility to speed up database deployment.
- Designed and built a license server for secure software distribution, and engineered protections that harden shipped applications against reverse engineering.
- Built cross-platform CI/CD and C/C++ build pipelines (CMake, Make) for Linux and Windows.

Software Engineer — RyoCurrency **10/2021 - 10/2022**

- Architected and deployed a full-stack application from planning to production, with a REST API in Django and PostgreSQL handling millions of records, containerized with Docker.
- Optimized search with the Manticore search engine, achieving a 200% performance improvement.
- Built an Apache log visualizer in Python (Sanic, Matplotlib) serving real-time traffic graphs behind Nginx.
- Built Python tooling on mitmproxy to analyze and decode mobile application traffic, scripts that encode telemetry in a binary format, and a Reddit post and comment collector.

PROJECTS AND OPEN SOURCE

LiSA — Desktop App to Stream and Download Anime and Manga (199 GitHub stars) GitHub

- Wrote the Python backend: a Starlette and Uvicorn API, with an Electron and React frontend, packaged for multiple operating systems with PyInstaller.
- Built an asyncio worker-pool downloader on aiohttp for HLS (m3u8) streams, with AES-128 segment decryption, resumable downloads tracked in SQLite, and live progress over WebSockets.
- Designed pluggable scraper and post-processing layers (abstract base classes) so new sources and output formats, such as manga-to-PDF, plug in without core changes.

chi_url — URL Shortener (FastAPI, Cassandra, Redis) GitHub

- Built a FastAPI service on Cassandra with a Redis cache that falls back to the database, JWT sessions, email verification, and per-URL stats, deployed with Gunicorn, Nginx, and Docker.

pyqtgraph — Contributor: fixed a bug that crashed applications on right-click in the scientific plotting and GUI library.

Omamori — Enterprise VPN Platform (500+ users): built a production multi-tenant SaaS in Go with gRPC, WireGuard, Chargebee billing, and AWS.

ACHIEVEMENTS

- Speaker at MumPy 2024: “Secure Distribution Strategy for Commercial Applications”
- Winner of the All-India Hackathon in Mumbai among 336 participants (2023)
- Semi-Finalist at the All-India Hackathon at Charotar University

EDUCATION

Bachelor of Technology in Information Technology **2023**
Savitribai Phule Pune University GPA: 9.0/10.0